

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics, and Economics

Honour School of Economics and Management

Honour School of History and Economics

MICROECONOMIC ANALYSIS

OUTLINE ANSWERS

TRINITY TERM 2023

Thursday 8 June, 14:30–17:30

Please start the answer to each question on a separate page.

There are ?? questions in this paper.

*Answer **four** questions.*

Candidates may use a calculator as specified by the Department of Economics.

Do not turn over until told that you may do so.

1. (a) Consider the following system of linear equations:

$$\begin{cases} x + y + cz = 1 \\ x + by + z = 2 \\ x + y + z = a \end{cases}$$

where $a, b, c \in \mathbb{R}$ are some parameters.

(i) Compute, for each possible values of a, b, c solutions of the system.

Solution: Using Gauss-Jordan elimination

$$\left(\begin{array}{ccc|c} 1 & 1 & c & 1 \\ 1 & b & 1 & 2 \\ 1 & 1 & 1 & a \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 1 & c & 1 \\ 0 & b-1 & 1-c & 1 \\ 0 & 0 & 1-c & a-1 \end{array} \right)$$

Thus,

A. If $c = 1$ and $a \neq 1$, then solution does not exist.

B. If $c = 1$ and $a = 1$, then the second equation simplifies to $(b-1)y = 1$ and thus

- if $b = 1$, then solution does not exist;
- if $b \neq 1$ then $y = \frac{1}{b-1}$ and the solution of the system is $(1 - \frac{1}{b-1} - cz, \frac{1}{b-1}, z)$ with $z \in \mathbb{R}$.

C. If $c \neq 1$ then $z = \frac{a-1}{1-c}$ and the second equation simplifies to $(b-1)y + (a-1) = 1$. Thus,

- if $b = 1$ and $a \neq 2$, then solution does not exist;
- if $b = 1$ and $a = 2$, then solution of the system is $(1 - y - \frac{a-1}{1-c}, y, \frac{a-1}{1-c})$ with $y \in \mathbb{R}$.
- Finally, if $b \neq 1$ then solution is $(1 - \frac{2-a}{b-1} - \frac{a-1}{1-c}, \frac{2-a}{b-1}, \frac{a-1}{1-c})$

Partial solutions should be marked proportionally to the number of cases considered (out of 6 total cases outlined above).

[30%]

(ii) For which values of b and c there is no value a such that the vector $\begin{pmatrix} 1 \\ 2 \\ a \end{pmatrix}$ is

spanned by vectors $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ b \\ 1 \end{pmatrix}, \begin{pmatrix} c \\ 1 \\ 1 \end{pmatrix}$?

Solution: As it follows from the previous answer the only situation when it can happen is if $b = c = 1$.

[10%]

- (iii) Compute the rank of matrix

$$A = \begin{pmatrix} 1 & 1 & c \\ 1 & b & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

depending on parameters b and c and explain how the rank you found is related to your answer to the previous question.

Solution: If $b, c \neq 1$ then $\text{rank } A = 3$ and any vector in $\mathbb{R}^3$ is spanned by columns of A . If either $b = 1$ or $c = 1$ then $\text{rank } A = 2$ and the vector is spanned only if it belongs to the same subspace (either $a = 2$ or $a = 1$ respectively). If $b = c = 1$ then $\text{rank } A = 1$, and vector $(1, 2, a)^T$ is never spanned by $(1, 1, 1)^T$. [10%]

- (iv) Compute the determinant of matrix A and explain how its value is related to the number of solutions of the system of linear equations you found in question (a.i).

Solution: $\det A = (b - 1)(c - 1)$. If $\det A \neq 0$, i.e. $b, c \neq 1$, then the system has a unique solution. Otherwise, the system has either none or infinitely many solutions. [10%]

- (b) A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *homogeneous of degree r* , if for all $t \in \mathbb{R}$ with $t \neq 0$,

$$f(t\mathbf{x}) = t^r f(\mathbf{x}).$$

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 function.

- (i) Show that if f is homogeneous of degree r , then all its partial derivatives are homogeneous of degree $r - 1$.

Solution: Take the partial derivative with respect to x_i in both sides of the equation:

$$t \frac{\partial f}{\partial x_i}(t\mathbf{x}) = t^r \frac{\partial f}{\partial x_i}(\mathbf{x}) \quad \Leftrightarrow \quad \frac{\partial f}{\partial x_i}(t\mathbf{x}) = t^{r-1} \frac{\partial f}{\partial x_i}(\mathbf{x})$$

[10%]

- (ii) Show that if f is homogeneous of degree r ,

$$\sum_{i=1}^n x_i \frac{\partial f}{\partial x_i}(\mathbf{x}) = r f(\mathbf{x}).$$

Solution: Take the partial derivative with respect to t in both sides of the equation:

$$\sum_{i=1}^n x_i \frac{\partial f}{\partial x_i}(t\mathbf{x}) = r t^{r-1} f(\mathbf{x}).$$

and evaluate at $t = 1$ to obtain the result.

[10%]

- (iii) Suppose that $n = 2$. By examining the Hessian, show that if f is homogeneous of degree 1, then f is either concave or convex on $\mathbb{R}^+ \times \mathbb{R}^+$.

Solution: By (i), both f_1 and f_2 are homogeneous of degree 0, hence by (ii)

$$x_1 f_{11} + x_2 f_{21} = 0 \quad \text{and} \quad x_1 f_{12} + x_2 f_{22} = 0$$

therefore, for any $x_1 > 0, x_2 > 0$, the Hessian can be written as:

$$H = \begin{pmatrix} -\frac{x_2}{x_1} f_{12} & f_{12} \\ f_{12} & -\frac{x_1}{x_2} f_{12} \end{pmatrix} = f_{12} \begin{pmatrix} -\frac{x_2}{x_1} & 1 \\ 1 & -\frac{x_1}{x_2} \end{pmatrix}$$

The determinant of H is 0 and the trace is $-f_{12}(\frac{x_2}{x_1} + \frac{x_1}{x_2})$ whose sign depends on f_{12} . If $f_{12} < 0$ then the function is convex (one eigenvalue is zero and the other is positive). If $f_{12} > 0$ the function is concave (one eigenvalue is zero and the other is negative) and if $f_{12} = 0$ the function is linear, so both concave and convex.

[20%]

2. A consumer has utility $u(x, y) = \alpha\sqrt{x} + (1 - \alpha)\sqrt{y}$ for the two available goods, where $\alpha \in (0, 1)$ is a preference parameter and $x, y \geq 0$. Her budget constraint is $x + y \leq 10$. Consuming each good takes time, so she also faces a time constraint: $2x + y \leq 12$.

(a) Show that the utility function is concave.

Solution: The Jacobian of the utility is $D_u = \begin{pmatrix} \frac{\alpha}{2}x^{-\frac{1}{2}} & \frac{(1-\alpha)}{2}y^{-\frac{1}{2}} \end{pmatrix}$. The hessian can be written as:

$$D_u^2 = \begin{pmatrix} -\frac{\alpha}{4}x^{-\frac{3}{2}} & 0 \\ 0 & -\frac{(1-\alpha)}{4}y^{-\frac{3}{2}} \end{pmatrix}$$

which is clearly negative definite: negative trace and positive determinant. Alternatively some students might have argue that the sum of two concave functions is concave, but then they need to show that $\sqrt{x}$ as a function of TWO variables is concave.

[10%]

(b) Explain why the non negativity constraints may be neglected.

Solution: The marginal utility of x ($MU_x = \frac{\alpha}{2}x^{-\frac{1}{2}}$) goes to infinity as x goes to 0. The same happens with the marginal utility of y ($MU_y = \frac{(1-\alpha)}{2}y^{-\frac{1}{2}}$). Therefore a positive amount of each good will be consumed.

[5%]

(c) Write down the Kuhn-Tucker first-order conditions with Lagrange multipliers λ_m and λ_t for the budget and the time constraint respectively. Comment on whether these conditions are (i) necessary, (ii) sufficient.

Solution: The Lagrangean is

$$L = \alpha\sqrt{x} + (1 - \alpha)\sqrt{y} - \lambda_m(x + y - 10) - \lambda_t(2x + y - 12)$$

The K-T conditions are:

$$\frac{\alpha}{2}x^{-\frac{1}{2}} - \lambda_m - 2\lambda_t = 0 \quad (1)$$

$$\frac{(1-\alpha)}{2}y^{-\frac{1}{2}} - \lambda_m - \lambda_t = 0 \quad (2)$$

$$\lambda_m(x + y - 10) = 0, \quad \lambda_m \geq 0, \quad x + y \leq 10 \quad (3)$$

$$\lambda_t(2x + y - 12) = 0, \quad \lambda_t \geq 0, \quad 2x + y \leq 12 \quad (4)$$

Given that the objective function is concave and the constraints are linear (and hence convex) we have a concave problem and the K-T conditions are sufficient. Moreover, since the constraint set has a non-empty interior ($x = 3, y = 3$ is an interior point), the conditions are also necessary. Alternatively, some students might say that the constraint qualification is satisfied (linear constraints) and therefore the conditions are necessary.

[20%]

(d) Find the consumer's optimal consumption as a function of α .

Solution: Note that at least one constraint will be binding. If none of them were, we could increase x or y and they bind and increase the utility.

- (i) Consider the case in which the two constraints bind. Then the solution is $x = 2$ and $y = 8$. In order to see when this would be the solution we need to check that the Lagrange multipliers are positive. Replacing $(x, y) = (2, 8)$ in (1) and (2) we get

$$\begin{aligned} 2\lambda_t + \lambda_m &= \frac{\alpha}{2\sqrt{2}} \\ \lambda_t + \lambda_m &= \frac{(1-\alpha)}{4\sqrt{2}} \end{aligned}$$

solving together:

$$\lambda_t = \frac{\alpha}{2\sqrt{2}} - \frac{(1-\alpha)}{4\sqrt{2}} \geq 0 \text{ if and only if } \alpha \geq \frac{1}{3}, \text{ and}$$

$$\lambda_m = -\frac{\alpha}{2\sqrt{2}} + \frac{(1-\alpha)}{2\sqrt{2}} \geq 0 \text{ if and only if } \alpha \leq \frac{1}{2}$$

Therefore $(x^*, y^*) = (2, 8)$ is the solution whenever $\frac{1}{3} \leq \alpha \leq \frac{1}{2}$.

- (ii) Consider the case in which only the budget constraint binds. Then $\lambda_t = 0$ by (4). (1) and (2) imply that $\lambda_m = \frac{\alpha}{2}x^{-\frac{1}{2}} = \frac{(1-\alpha)}{2}y^{-\frac{1}{2}} > 0$. In particular: $\alpha^2 y = (1-\alpha)^2 x$, and replacing in the budget constraint we get:

$$(x^*, y^*) = \left(\frac{10\alpha^2}{\alpha^2 + (1-\alpha)^2}, \frac{10(1-\alpha)^2}{\alpha^2 + (1-\alpha)^2} \right).$$

This is a solution as long as the time constraint is satisfied: $2x^* + y^* \leq 12$ as long as $20\alpha^2 + 10(1-\alpha)^2 \leq 12\alpha^2 + 12(1-\alpha)^2$ which is equivalent to $\alpha \leq \frac{1}{3}$.

- (iii) Finally, consider the case in which only the time constraint binds. Then $\lambda_m = 0$ and (1) and (2) imply: $\lambda_t = \frac{\alpha}{4}x^{-\frac{1}{2}} = \frac{(1-\alpha)}{2}y^{-\frac{1}{2}} > 0$. In particular $4(1-\alpha)^2 x = \alpha^2 y$ and replacing in the time constraint we get:

$$(x^*, y^*) = \left(\frac{6\alpha^2}{\alpha^2 + 2(1-\alpha)^2}, \frac{24(1-\alpha)^2}{\alpha^2 + 2(1-\alpha)^2} \right).$$

This is a solution as long as the budget constraint is satisfied: $x^* + y^* \leq 10$ which is equivalent to $\alpha \geq \frac{1}{2}$.

Hence, the optimal consumption bundle is:

$$(x^*, y^*) = \begin{cases} \left(\frac{10\alpha^2}{\alpha^2 + (1-\alpha)^2}, \frac{10(1-\alpha)^2}{\alpha^2 + (1-\alpha)^2} \right) & \text{if } \alpha \leq \frac{1}{3} \\ (2, 8) & \text{if } \frac{1}{3} \leq \alpha \leq \frac{1}{2} \\ \left(\frac{6\alpha^2}{\alpha^2 + 2(1-\alpha)^2}, \frac{24(1-\alpha)^2}{\alpha^2 + 2(1-\alpha)^2} \right) & \text{if } \alpha \geq \frac{1}{2} \end{cases}$$

[40%]

- (e) When $\alpha = \frac{1}{4}$, check that the solution to the Kuhn-Tucker first order conditions is $(x^*, y^*, \lambda_m^*, \lambda_t^*) = (1, 9, \frac{1}{8}, 0)$.

Solution: Immediate from the solution above. (this is only provided so that those stuck with the question above can continue).

- (i) When $\alpha = \frac{1}{4}$, how does the consumer's utility change if there is a marginal increase in income?

Solution: The marginal utility of income corresponds to the lagrange multiplier for the budget constrain. In this case $\lambda_m = \frac{1}{8}$.

- (ii) When $\alpha = \frac{1}{4}$, how does the consumer's utility change if there is a marginal increase in time?

Solution: In this case the time constraint is not binding! Therefore increasing time marginally does not affect the consumers utility. This can also be seen by the lagrange multiplier of the time constrain which is $\lambda_t = 0$.

- (iii) How does the consumer's utility change when α marginally increases from $\frac{1}{4}$?

Solution: By the Envelop Theorem, we can ignore the changes through the optimal bundle and focus on the direct effect. Hence the change in utility is $\sqrt{x^*} - \sqrt{y^*} = 1 - 3 = -2$.

- (iv) Without further calculations and without using your explicit solution to part (d), explain how you could compute the change in the optimal consumption bundle when α marginally increases.

Solution: When $\alpha = \frac{1}{4}$, the budget constraint binds and the time constraint is slack ($2 + 9 < 12$). Hence $(\alpha, x^*, y^*, \lambda_m^*) = (\frac{1}{4}, 1, 9, \frac{1}{8})$ solves the following system of equations given by the K-T conditions:

$$\begin{aligned} F_1(\alpha, x, y, \lambda_m) &= \frac{\alpha}{2}x^{-\frac{1}{2}} - \lambda_m = 0 \\ F_2(\alpha, x, y, \lambda_m) &= \frac{(1-\alpha)}{2}y^{-\frac{1}{2}} - \lambda_m = 0 \\ F_3(\alpha, x, y, \lambda_m) &= x + y - 10 = 0 \end{aligned}$$

We can check that $D_{x,y,\lambda_m}\mathbf{F}(\frac{1}{4}, 1, 9, \frac{1}{8})$ is invertible and apply the implicit function theorem ($\mathbf{F}$ is C^1) to find that:

$$\begin{pmatrix} \frac{\partial x^*}{\partial \alpha}(\frac{1}{4}) & \frac{\partial y^*}{\partial \alpha}(\frac{1}{4}) & \frac{\partial \lambda_m^*}{\partial \alpha}(\frac{1}{4}) \end{pmatrix} = -D_{x,y,\lambda_m}\mathbf{F}^{-1}(\frac{1}{4}, 1, 9, \frac{1}{8})D_{\alpha}\mathbf{F}(\frac{1}{4}, 1, 9, \frac{1}{8}). \quad [25\%]$$

3. Consider lotteries over three fixed monetary outcomes with values $x_1 < x_2 < x_3$. A lottery is given by a probability distribution (p_1, p_2, p_3) where $p_i = \Pr[x_i] \geq 0$ for $i = 1, 2, 3$ and $p_1 + p_2 + p_3 = 1$.

- (a) Ann is an expected utility maximizer with strictly monotonic preferences. She is indifferent between lotteries $(p_1, p_2, p_3) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ and $(q_1, q_2, q_3) = (\frac{2}{3}, 0, \frac{1}{3})$. How does she compare “ x_2 for sure” to either of these two lotteries? Can you tell whether or not she is risk averse? Explain.

Solution: Without loss of generality $u(x_1) = 0$, $u(x_3) = 1$, and $u(x_2) = u \in (0, 1)$. The EU of $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ and $(\frac{2}{3}, 0, \frac{1}{3})$ being equal yields $\frac{1}{4}u + \frac{1}{4} = \frac{1}{3}$, hence $u = \frac{1}{3}$. This is also the expected utility of “ x_2 for sure”, therefore the agent is indifferent between all three. We cannot determine the agent’s risk preferences. We know that the certainty equivalent of, e.g., $(\frac{2}{3}, 0, \frac{1}{3})$ is x_2 . Ann is (strictly) risk averse iff the expected value of the former lottery (strictly) exceeds the certainty equivalent, i.e., $\frac{2}{3}x_1 + \frac{1}{3}x_3 \geq (>)x_2$.

[30%]

- (b) In this part only, let $x_1 = 1$, $x_2 = 2$, $x_3 = 3$. Bob is a risk neutral expected utility maximizer. His original final-wealth position is x_2 , but he can increase the probability of outcome x_3 to any $p_3 \in [0, \frac{1}{2}]$ at the “cost” of increasing the probability of outcome $x_1 = 1$ to $p_1 = 2(p_3)^2 \in [0, \frac{1}{2}]$ as well. (The probabilities must add up to one, so p_2 decreases by the appropriate amount.) What lottery will he choose? Does his choice first- or second-order stochastically dominate any of the other lotteries he could have chosen?

Solution: If Bob chooses $p_3 \in [0, \frac{1}{2}]$ then the expected value of final wealth becomes $EV = (2p_3^2) \cdot 1 + (1 - 2p_3^2 - p_3) \cdot 2 + p_3 \cdot 3 = 2 - 2p_3^2 + p_3$. This is strictly concave hence maximized where the FOC holds, $-4p_3 + 1 = 0$, i.e., $p_3 = \frac{1}{4}$. Bob chooses $p^* = (p_1^*, p_2^*, p_3^*) = (\frac{1}{8}, \frac{5}{8}, \frac{1}{4})$. The probabilities of the lowest and highest outcomes both increase in p_3 , therefore p^* does not first-order stochastically dominate any other lottery available to Bob. Second-order stochastic dominance does not apply either as no other lottery available to Bob yields the same expected value (EV is uniquely maximized at p^*).

[30%]

- (c) Cecil’s strict preference relation $\succ_C$ is defined as follows. If the expected value of lottery $p = (p_1, p_2, p_3)$ strictly exceeds that of $q = (q_1, q_2, q_3)$ then $p \succ_C q$; if the expected values are equal then $p \succ_C q$ if and only if the variance of p is strictly smaller than that of q . Cecil’s weak preference $\succeq_C$ is derived from $\succ_C$ as usual: $p \succeq_C q$ if and only if $q \succ_C p$ does not hold. Are Cecil’s preferences rational?

Solution: First, $\succeq_C$ is complete because for any two lotteries p and q either their

EV's are different in which case they are strictly (hence also weakly) ranked, or their EV's are equal, in which case either their variances are different implying again strict hence weak ranking, or their EV's and variances are equal and then $p \succeq_C q$ and $q \succeq_C p$ both hold. Second, $\succeq_C$ is transitive because $p \succeq_C q$ and $q \succeq_C r$ imply $EV[p] \geq EV[q] \geq EV[r]$, hence $r \succ_C p$ cannot hold, hence $p \succeq_C r$. Therefore $\succeq_C$ is rational.

[15%]

- (d) Continue with the assumptions in part (c). Does Cecil prefer a lottery that first- or second-order stochastically dominates another? Can Cecil's preferences be represented by an expected utility function? Either provide a representation or explain which axiom(s) fail.

Solution: Cecil prefers p to q if $p \succ_{\text{FOSD}} q$ because the latter implies $EV[p] > EV[q]$. Cecil also prefers p to q if $p \succ_{\text{SOSD}} q$ because the latter means $EV[p] = EV[q]$ and that q is a mean-preserving spread of p , hence $\text{Var}[p] < \text{Var}[q]$. However, Cecil's preferences cannot be represented by a utility function (let alone vNM expected utility) on the set of lotteries, because $\succ_C$ is not continuous; in fact it is lexicographic (Cecil's first priority is EV, second priority is Var). An example why/how continuity fails deserves a first-class mark. Suppose $x_1 = 1$, $x_2 = 2$, $x_3 = 3$ and take a sequence of lotteries $p^{(n)} = (\frac{1}{2} - \frac{1}{n}, 0, \frac{1}{2} + \frac{1}{n})$. For all $n = 1, 2, \dots$ the EV of these lotteries strictly exceeds $x_2 = 2$, hence $p^{(n)} \succ_C q = (0, 1, 0)$. However, $p^{(n)}$ converges to the lottery $(\frac{1}{2}, 0, \frac{1}{2})$, which has the same EV as " x_2 for sure" but a greater (positive) variance, so in the limit strict preference is reversed. A really-really good solution might note that the other conditions (rational preferences over lotteries and the independence axiom) are not violated.

[25%]

4. (a) Let weak preference relation $\succeq$ be defined over dated monetary payoffs (z, t) corresponding to amount $z \in \mathbb{R}_+$ received in period $t = 0, 1, \dots, T$. Under what conditions does $\succeq$ have a discounted utility representation with utility function $u(z)$ and discount factor δ such that $(z, t) \succeq (x, s) \Leftrightarrow \delta^t u(z) \geq \delta^s u(x)$? List and interpret the conditions, do not give a proof.

Solution: If $\succeq$ is rational (complete and transitive) and continuous (weakly-preferred sets are closed) then a utility function $U(z, t)$ that is continuous in z exists by Debreu's Theorem. Monotonicity of $\succeq$ in z implies $U(z, t)$ is increasing in z as well. The crucial additional axiom for discounted utility representation is stationarity: $(z, t) \succeq (x, s) \Leftrightarrow (z, t + r) \succeq (x, s + r)$. Allocate 15 points to the discussion of increasing utility representation and 10 points for stationarity.

[25%]

In what follows there are two periods, $t = 0, 1$. An agent has initial wealth w_0 in period 0 and w_1 in period 1. In period 0 the agent chooses n and pays $\mathcal{L}4n$; in period 1 the agent gets a monetary payment equal to $\mathcal{L}100X_n$ where X_n is a random variable equal to the maximum of n independent draws from $[0, 1]$ according to the uniform distribution.

- (b) Recall that the probability that a single uniform draw from $[0, 1]$ is at or below $x \in [0, 1]$ is exactly x . Explain why the cumulative distribution function (cdf) of X_n is $F_n(x) = x^n$. What is the corresponding probability density function, $f_n(x)$?

Solution: $F_n(x) = \Pr[X_n \leq x] = \Pr[n \text{ iid uniform draws in } [0, 1] \text{ all below } x] = x^n$ because the probability that each draw is below x is x . The corresponding pdf is $f_n(x) = nx^{n-1}$.

[10%]

- (c) Suppose that Chris is risk neutral and maximizes his total final monetary wealth in periods $t = 0, 1$ without discounting. His initial wealth levels w_0 and w_1 are sufficiently large. He chooses n and pays $4n$ at $t = 0$, then receives $100X_n$ at $t = 1$. Write down his objective function and compute what n he will choose. Does his choice depend on w_0 and w_1 ?

Solution: Initial wealth levels won't matter: as Chris is risk neutral, w_0 and w_1 don't affect his marginal utility of n at $t = 0, 1$. Chris' total gain relative to $w_0 + w_1$ is

$$\begin{aligned} -4n + \int_0^1 100x(nx^{n-1}) dx &= -4n + n \int_0^1 100x^n dx \\ &= \frac{n}{n+1} 100 - 4n \end{aligned}$$

Candidates may assert (know) without calculation that the maximum of n independent uniform draws from $[0, 1]$ is on average $n/(n+1)$ and get the above expected

payoff without integration. Chris's payoff is concave in n , so we take the FOC to find the maximum in n ,

$$\frac{100}{(n+1)^2} - 4 = 0 \Leftrightarrow 25 = (n+1)^2 \Rightarrow n = 4 \text{ (keep positive root)}$$

Luckily this is an integer ;-), so Chris chooses $n = 4$ as long as $w_0 \geq 16$ otherwise the problem will be constrained by w_0 .

[30%]

- (d) Dina's initial wealth levels are $w_0 = w_1 = 40$. She chooses n at $t = 0$ and pays $4n$, then receives $100X_n$ at $t = 1$. Her preferences are additively separable over the two periods and can be represented by the utility function $\ln(w_0 - 4n) + (w_1 + 100X_n)$. What is her optimal choice of n ?

Solution: The FOC of Dina's problem is $\frac{d}{dn} \ln(w_0 - 4n) + 100 \frac{d}{dn} \left(\frac{n}{n+1} \right) = 0$, i.e.,

$$\frac{4}{40 - 4n} = \frac{100}{(n+1)^2} \Leftrightarrow n^2 + 102n + 1 - 1000 = 0.$$

The FOC is sufficient as her objective is strictly concave. The solution is $n = 9$, hence this is Dina's choice of n .

[20%]

- (e) How does Dina's optimal choice of n depend on her initial wealth levels, w_0 and w_1 ? Explain.

Solution: As Dina is risk neutral at $t = 1$ her initial wealth w_1 makes no difference for her marginal utility at $t = 1$. However, her marginal utility at $t = 0$ is decreasing in w_0 , therefore she feels the loss $4n$ less keenly at higher w_0 , therefore she chooses a larger n with a larger w_0 , i.e., the optimal n is increasing in w_0 . A good solution may establish or verify this by differentiation (comparative statics methods).

[15%]

5. A company offers a summer internship to a student. The contract specifies the total wage, w , and the duration of the internship, d , in days. The company earns a profit of $\pi(w, d) = 2d^{1/2} - w$ when the internship lasts d days and the total wage is w . The student's utility if she accepts the internship is $u(w, d) = w - d\theta$ where θ represents the disutility of spending a day at work. There are two types of students. Those with low disutility of working, θ_L , and those with high disutility of working, θ_H . Assume that $0 < \theta_L < \theta_H$ and denote by λ the probability the student has a low disutility. If the student rejects the offer, her reservation utility depends on her type, it is $\bar{u}_H = 0$ if $\theta = \theta_H$ and $\bar{u}_L \geq 0$ if $\theta = \theta_L$.

(a) Suppose that θ is publicly known at the time of signing the contract.

(i) Write the company's problem.

Solution: The company's problem is: $\max_{w_i, d_i} \{2d_i^{1/2} - w_i, 0\}$ subject to the participation constraint $w_i - \theta_i \geq \bar{u}_i$, as long as the profits are positive. Note that the company observes the types, there is a different problem for each type.

[5%]

(ii) Find the first best internship (w, d) for each θ . For which values of $\bar{u}_L$, the company offers an internship to each type of student?

Solution: The participation constraint will bind, otherwise the company can decrease w_i or increase d_i while satisfying the constraint and therefore increase its profits. Hence $w_i = \bar{u}_i + d_i\theta_i$. Replacing in the profits and taking first order conditions (second order conditions are satisfied by concavity):

$$d_i^{-1/2} - \theta_i = 0 \quad \Leftrightarrow \quad d_i = \left(\frac{1}{\theta_i}\right)^2, \quad w_i = \bar{u}_i + \frac{1}{\theta_i}$$

So the two first best contracts are $(w_H^*, d_H^*) = (\frac{1}{\theta_H}, \frac{1}{\theta_H^2})$ and $(w_L^*, d_L^*) = (\bar{u}_L + \frac{1}{\theta_L}, \frac{1}{\theta_L^2})$, as long as $\bar{u}_L \leq \frac{1}{\theta_L}$ so that profits are positive.

[15%]

(iii) Sketch this situation on a diagram with d on the horizontal axis and w on the vertical axis. Your diagram should include the first best internships the indifference curves for both types of student and the iso-profit curves for the company.

Solution Indifference curves are straight lines with slope θ_H and θ_L . The participation constraint for the high type passes through the origin. For the low type it has intercept $\bar{u}_L \geq 0$. The iso-profit lines are concave curves (parallel shifts given the quasi-linearity of the profit function). The first best should be the points at which the iso-profit curves are tangent to the participation constraints.

[10%]

From now onwards, suppose that at the time of signing the contract, the student knows her type θ , but the company does not. The company offers two internships (w_L, d_L) , (w_H, d_H) and allows the student to select the one she prefers.

- (b) Assuming that the firm wants both types of student to participate in the contract, write the new problem the company faces. For which values of $\bar{u}_L$ does the company reach the first best?

Solution: The company's problem is:

$$\begin{aligned} \max_{w_H, d_H, w_L, d_L} \quad & \lambda(2d_L^{1/2} - w_L) + (1 - \lambda)(2d_H^{1/2} - w_H) \\ \text{s.t.} \quad & w_L - d_L\theta_L \geq \bar{u}_L \quad (PC_L) \\ & w_H - d_H\theta_H \geq 0 \quad (PC_H) \\ & w_L - d_L\theta_L \geq w_H - d_H\theta_L \quad (IC_L) \\ & w_H - d_H\theta_H \geq w_L - d_L\theta_H \quad (IC_H) \end{aligned}$$

For the First best to be achievable they need to be incentive compatible, so H prefers (w_H^*, d_H^*) to (w_L^*, d_L^*) and L prefers (w_L^*, d_L^*) to (w_H^*, d_H^*) . These constraints can be written as:

$$\begin{aligned} w_H^* - d_H^*\theta_H \geq w_L^* - d_L^*\theta_H & \Leftrightarrow \bar{u}_L \leq \frac{\theta_H - \theta_L}{\theta_L^2} \\ w_L^* - d_L^*\theta_L \geq w_H^* - d_H^*\theta_L & \Leftrightarrow \bar{u}_L \geq \frac{\theta_H - \theta_L}{\theta_H^2} \end{aligned}$$

Hence, the first best is achievable when

$$\frac{\theta_H - \theta_L}{\theta_H^2} \leq \bar{u}_L \leq \frac{\theta_H - \theta_L}{\theta_L^2}$$

[10%]

- (c) Suppose that $\bar{u}_L = 0$.

- (i) Show graphically or otherwise that in the optimal contract the student with high disutility is indifferent between taking the internship or not.

Solution: First show that if PC_H and IC_L are satisfied then PC_L will be as well:

$$w_L - d_L\theta_L \geq w_H - d_H\theta_L \geq w_H - d_H\theta_H \geq 0$$

We can therefore ignore PC_L as long as PC_H and IC_L are satisfied. Suppose PC_H is not binding, then increase d_H and d_L by the same amount until PC_H binds. Note that the IC's constraint are unchanged. If shown graphically, the argument should be clear.

[10%]

- (ii) Show graphically or otherwise that in the optimal contract the student with low disutility is indifferent between the two internships.

Solution: Suppose that IC_L is not binding, then increase d_L until it binds. Note that this doesn't affect PC_H , and IC_H is relaxed. As we argued above, PC_H is not a concern.

[10%]

- (iii) Show graphically or otherwise that the length of the internship intended for the

student with low disutility is the same as in the first best internship of part (a.ii).

Solution: Given IC_L is binding, the optimal point is where the iso-profit is tangent to the IC_L constraint (which happens at the same level of d_L because of quasi-linearity.) Note that at this point the IC_H is satisfied, so no risk of high types choosing this contract. Algebraically, we can use the two binding constraints to write $w_H = d_H\theta_H$ and $w_L = d_H\theta_H + \theta_L(d_L - d_H)$, and replace them in the objective function. The problem becomes (ignoring IC_H):

$$\max_{d_H, d_L} \lambda(2d^{1/2} - d_H\theta_H - \theta_L(d_L - d_H)) + (1 - \lambda)(2d_H^{1/2} - d_H\theta_H)$$

First order conditions leads to:

$$\begin{aligned} \lambda(d_L^{-1/2} - \theta_L) &= 0 \\ \lambda(\theta_L - \theta_H) + (1 - \lambda)(d_H^{-1/2} - \theta_H) &= 0 \end{aligned}$$

So $d_L = \frac{1}{\theta_H^2}$ as in the first best contract. (for a complete algebraic answer, the student should check that the IC_H is satisfied in the relaxed solution (which it is). Do not penalise too much if they don't.

[10%]

- (iv) Provide an intuition of how the length of the internship chosen by the student with high disutility compares to the first best.

Solution: Only an intuition is required. Graphically it is easy to see that $d_H > d_H^*$ will never be optimal, as $d_L = d_L^*$ and w_L increases is d_H (we need to make L indifferent), so the profits of the company decrease in $d_H > d_H^*$. In fact we can show that $d_H < d_H^*$ as a marginal decrease in d_H from d_H^* has zero first order effect on the profits derived from H and a discrete gain in the profits extracted from H .

[10%]

- (d) Without further computations, explain how the optimal contract changes as $\bar{u}_L$ increases.

Solution: In the optimal solution the PC_L is slack, so the solution will not change until $\bar{u}_L$ increases enough to make PC_L bind. Once such a level is reached, then PC_L will need to bind (by increasing w_L -bad for the company) and this allows d_H and w_H to increase accordingly (good for the company), until (w_H^*, d_H^*) is reached ($\bar{u}_L = \frac{\theta_H - \theta_L}{\theta_H^2}$). As $\bar{u}_L$ increases further the solution coincides with the first best until either $\bar{u}_L = \frac{\theta_H - \theta_L}{\theta_L^2}$ or $\bar{u}_L = \frac{1}{\theta_L}$ whatever is smaller. If $\frac{1}{\theta_L} < \frac{\theta_H - \theta_L}{\theta_L^2}$, then $\bar{u}_L > \frac{1}{\theta_L}$ only H types will be offered the contract (w_H^*, d_H^*) . If $\frac{1}{\theta_L} > \bar{u}_L > \frac{\theta_H - \theta_L}{\theta_L^2}$, then the relevant binding constraints become IC_H , and PC_L . ($d_H = d_H^*$, and $d_L < d_L^*$). I don't expect students to be able to specify all these paths, but if one attempts many of these paths, be generous with your marks.

[20%]

6. Consider a seller who has a stock of product s_0 to be sold across T periods. The product is non-perishable and costless to store. In each period $t = 0, 1, \dots, T-1$ there is a group of consumers in the market with inverse demand function $p_t = q_t^{\gamma-1}$, where q_t is the amount of good sold in the market and $\gamma \in (0, 1)$. The seller maximizes its discounted profits subject to sales in all periods not exceeding the initial stock:

$$\max_{q_t > 0} \sum_{t=0}^{T-1} \delta^t p_t q_t \quad \text{s.t.} \quad \sum_{t=0}^{\infty} q_t \leq s_0$$

where $\delta < 1$ is the discount factor.

- (a) Define the state variable and the control variable for this problem. Write the instantaneous payoff of the seller as a function of the control variable.

Solution: State variable is s_t , control variable is q_t and instantaneous payoff is $\pi_t = p_t q_t = q_t^\gamma$.

[10%]

- (b) Suppose that $T = 2$.

- (i) What is the optimal sales plan for both periods? What is the total discounted profit?

Solution: For $T = 2$ the problem is solved using backward induction. If the second period is reached with stock s_1 , then $q_1 = s_1$. Then, overall profit is given by

$$V(s_0) = q_0^\gamma + \delta q_1^\gamma = (s_0 - s_1)^\gamma + \delta s_1^\gamma.$$

First order condition gives:

$$(s_0 - s_1)^{\gamma-1} = \delta s_1^{\gamma-1} \quad \Rightarrow \quad s_1 = \frac{1}{1 + \delta^{\frac{1}{\gamma-1}}} s_0.$$

Thus, total profit is given by:

$$V(s_0) = \left[\left(\frac{\delta^{\frac{1}{\gamma-1}}}{1 + \delta^{\frac{1}{\gamma-1}}} \right)^\gamma + \delta \left(\frac{1}{1 + \delta^{\frac{1}{\gamma-1}}} \right)^\gamma \right] s_0^\gamma = \frac{\delta}{\left(1 + \delta^{\frac{1}{\gamma-1}} \right)^{\gamma-1}} s_0^\gamma$$

[15%]

- (ii) What happens with q_0 and q_1 as $\delta \rightarrow 1$? Give an intuitive explanation for this result.

Solution: If $\delta = 1$ then $s_1 = s_0/2$, thus the same quantity is sold in any period. This is intuitive: there is no discounting, but the marginal revenue is decreasing (and total revenue is concave), so equal split gives the highest total revenue.

[10%]

- (c) Now suppose that $T = \infty$.

- (i) Rewrite the seller's problem as a sequence problem and set up the corresponding Bellman equation.

Solution: Note, that $s_t = q_t + s_{t+1}$. Thus, we can rewrite the problem as

$$V(s_0) = \max_{s_t: s_{t+1} < s_t} \sum_{t=0}^{\infty} \delta^t (s_t - s_{t+1})^\gamma.$$

Corresponding Bellman equation is

$$V(s_t) = \max_{s_t: 0 \leq s_{t+1} < s_t} [(s_t - s_{t+1})^\gamma + \delta V(s_{t+1})].$$

[10%]

- (ii) Solve the Bellman equation. [Hint: try to use the functional form of the value function from $T = 2$ case].

Solution: Conjecture that $V(s_t) = C s_t^\gamma$, where C is some constant. Then the Bellman equation can be rewritten as

$$C(s_t)^\gamma = \max_{s_t: 0 \leq s_{t+1} < s_t} ((s_t - s_{t+1})^\gamma + \delta C s_{t+1}^\gamma).$$

First order condition gives

$$(s_t - s_{t+1})^{\gamma-1} = \delta C s_{t+1}^{\gamma-1} \Rightarrow s_{t+1} = \frac{1}{1 + (\delta C)^{\frac{1}{\gamma-1}}} s_t.$$

Plugging it back into the Bellman equation gives

$$C s_t^\gamma = \left(1 + \frac{1}{1 + (\delta C)^{\frac{1}{\gamma-1}}}\right)^\gamma s_t^\gamma + C \delta \left(\frac{1}{1 + (\delta C)^{\frac{1}{\gamma-1}}}\right)^\gamma s_t^\gamma = \frac{C \delta}{\left(1 + (\delta C)^{\frac{1}{\gamma-1}}\right)^{\gamma-1}} s_t^\gamma.$$

Therefore,

$$\delta^{\frac{1}{\gamma-1}} = 1 + C^{\frac{1}{\gamma-1}} \delta^{\frac{1}{\gamma-1}} \Rightarrow C = \frac{\left(\delta^{\frac{1}{\gamma-1}} - 1\right)^{\gamma-1}}{\delta} = \left(1 - \delta^{\frac{1}{1-\gamma}}\right)^{\gamma-1}.$$

Therefore,

$$s_{t+1} = \delta^{\frac{1}{1-\gamma}} s_t, \quad V(s_0) = \left(1 - \delta^{\frac{1}{1-\gamma}}\right)^{\gamma-1} s_0^\gamma.$$

[45%]

- (iii) What happens with q_t as $\delta \rightarrow 1$? Give an intuitive explanation for this result.

Solution: Note, that $q_{t+1} = s_{t+1} - s_t = (1 - \delta^{\frac{1}{1-\gamma}}) s_t$. Thus, if $\delta \rightarrow 1$, the firm will sell smaller and smaller amount in each period. This is a consequence of decreasing marginal value: the firm prefers to make a large number of smaller high-margin sells to selling quickly at a smaller margin.

[10%]